1️⃣ What is a Jenkins Pipeline?

A **Jenkins Pipeline** is a **suite of automated processes that define how software is built, tested, and deployed** in a Continuous Integration / Continuous Delivery (CI/CD) workflow.

It allows developers to **define the entire build process as code** using a file called **Jenkinsfile**.

💡 In simple terms:
A Jenkins Pipeline describes **all the steps required to build and deliver an application automatically.**

Typical stages in a pipeline include:

- **Build** – Compile the code
- **Test** – Run automated tests
- **Package** – Create deployable artifacts
- **Deploy** – Deploy to staging or production

Example pipeline flow:

Code Commit → Build → Test → Package → Deploy

Benefits of Jenkins Pipeline:

- Pipeline **as Code**
- Easy to **version control**
- Supports **complex workflows**
- Enables **CI/CD automation**
- Better **visualization of build stages**

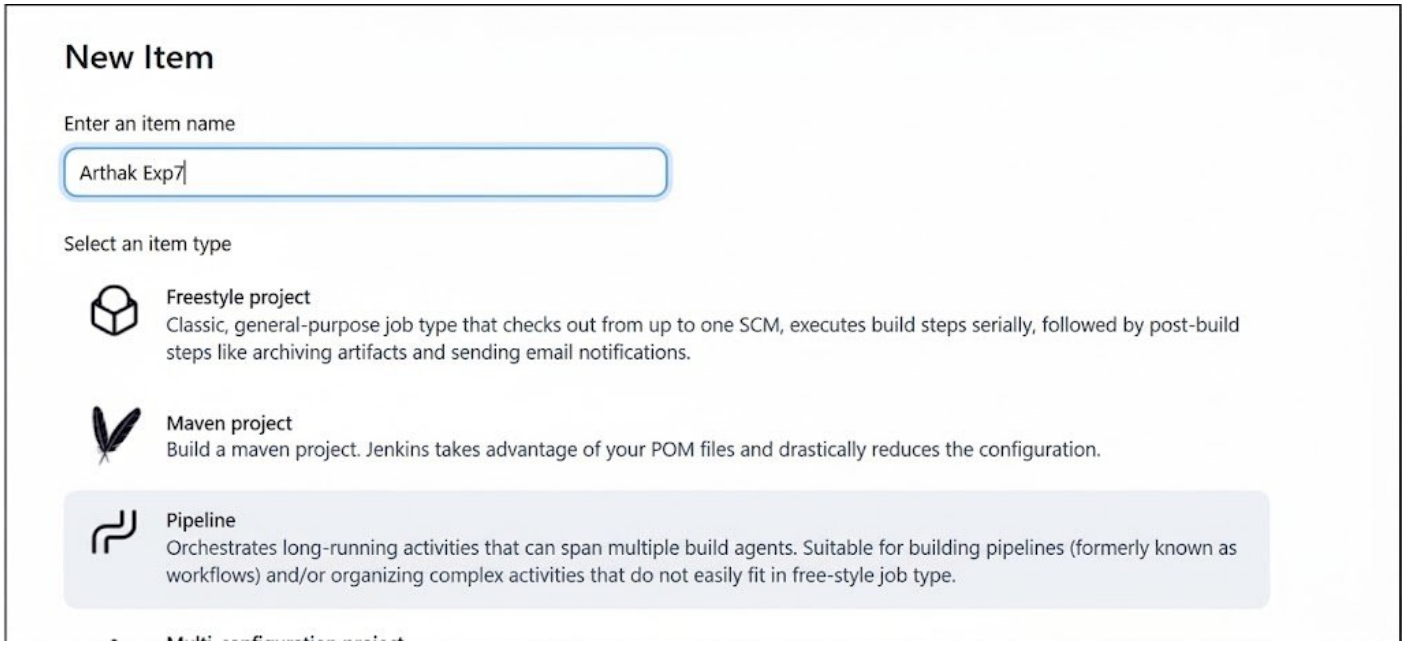
Different Ways to Write a Jenkins Pipeline

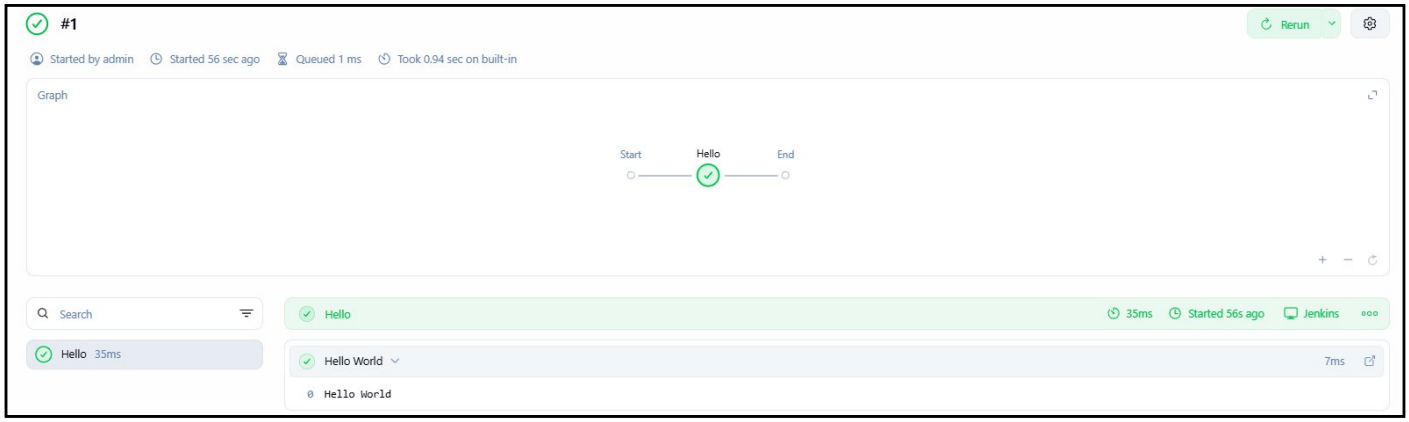
In **Jenkins**, pipelines can be written in **two main ways**: **Declarative Pipeline** and **Scripted Pipeline**. Both methods define the steps required to build, test, and deploy an application automatically.

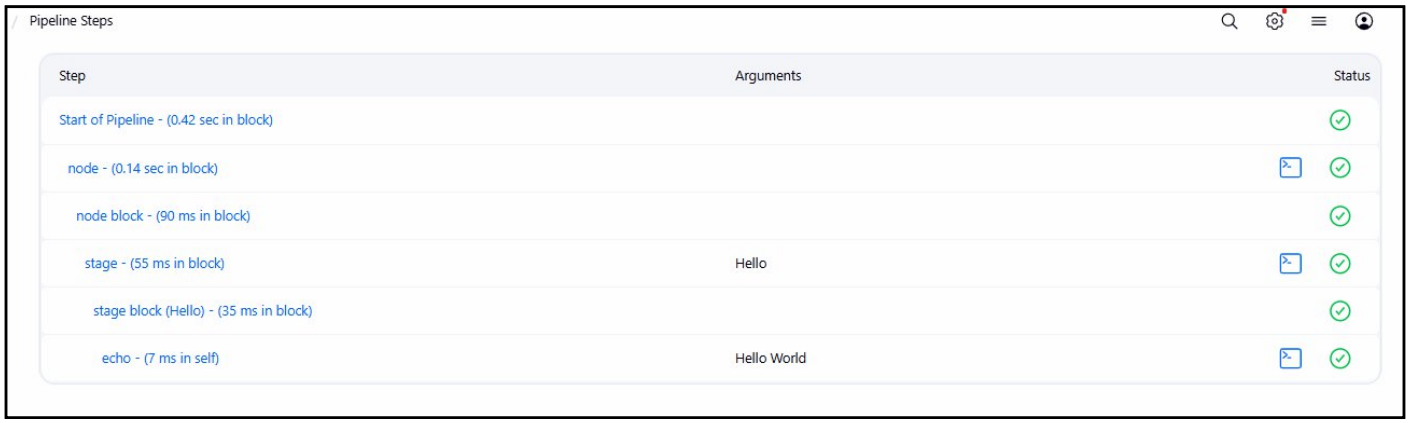
1. Declarative Pipeline:
Declarative Pipeline uses a simple and structured syntax. It is the most commonly used and recommended approach because it is easy to read, write, and maintain. It uses predefined sections such as pipeline, agent, stages, and steps to define the workflow. This format helps developers clearly organize different stages like build, test, and deploy.

2. Scripted Pipeline:
Scripted Pipeline is written using **Groovy** scripting language. It provides more flexibility and control over the pipeline logic. Developers can write complex conditions, loops, and custom logic using Groovy code. However, it is more complex compared to the Declarative Pipeline.

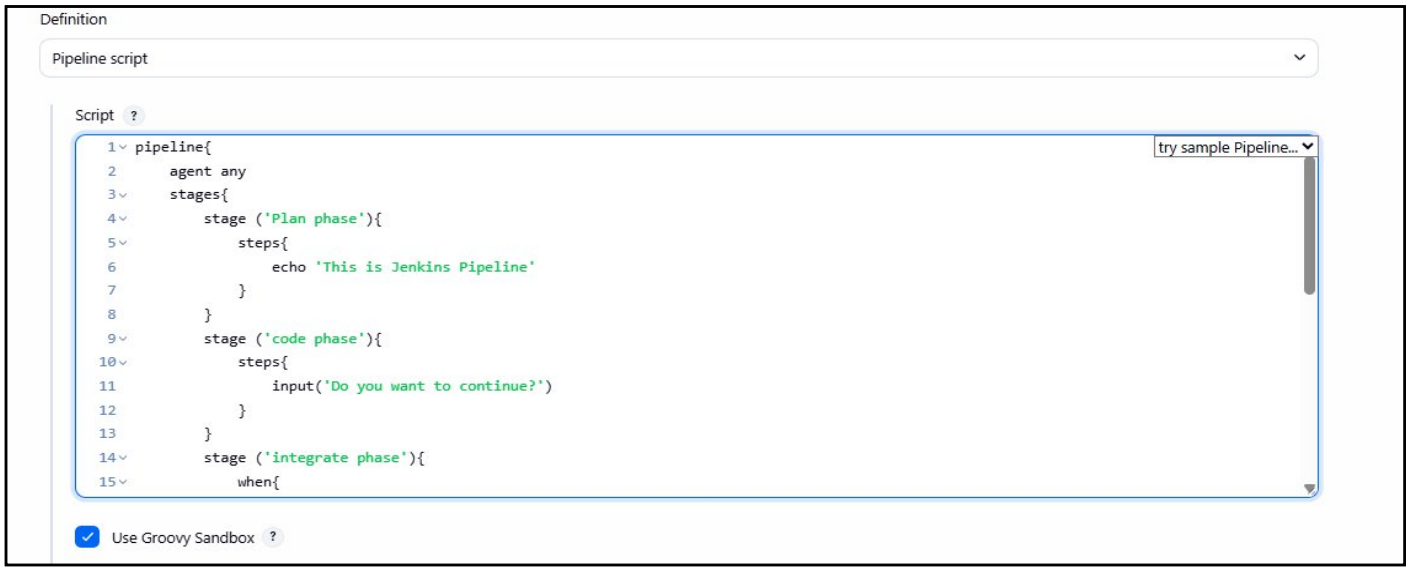
1. Create and build Pipeline Hello World with pipeline script



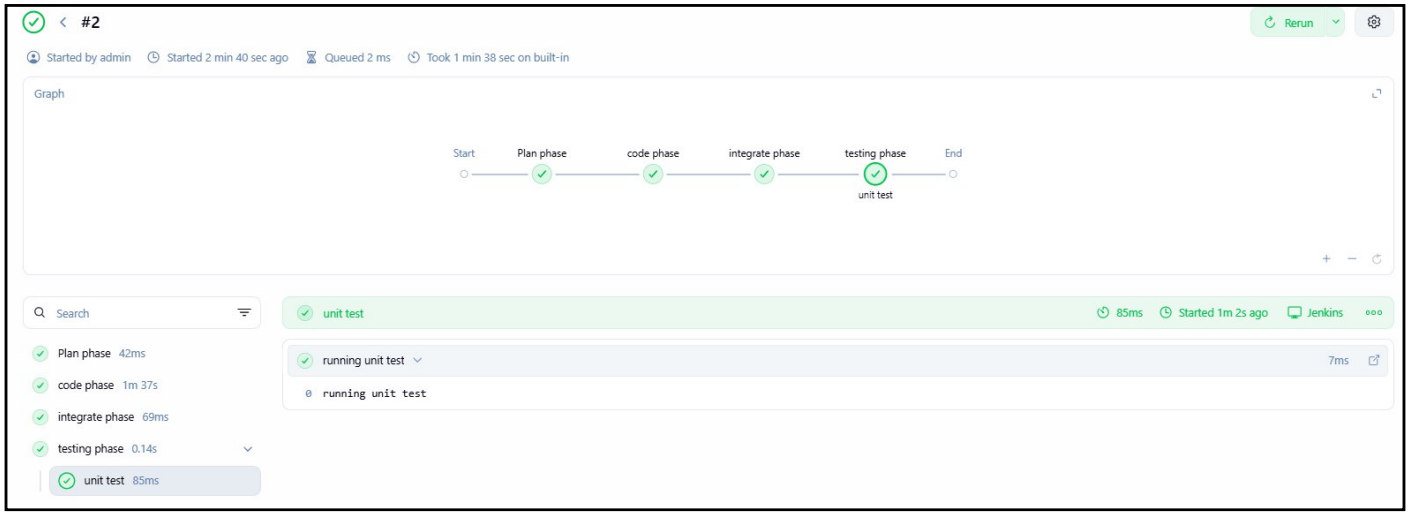


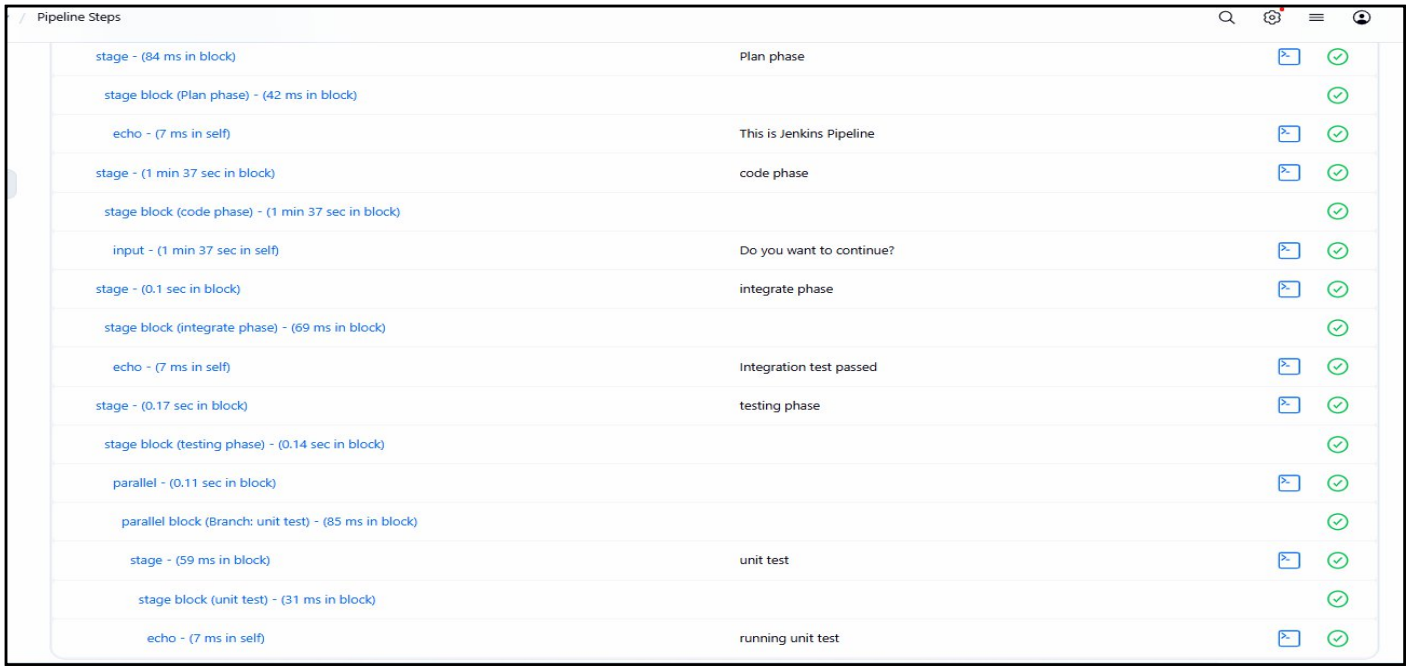


2. Pipeline script Project with your own pipeline script

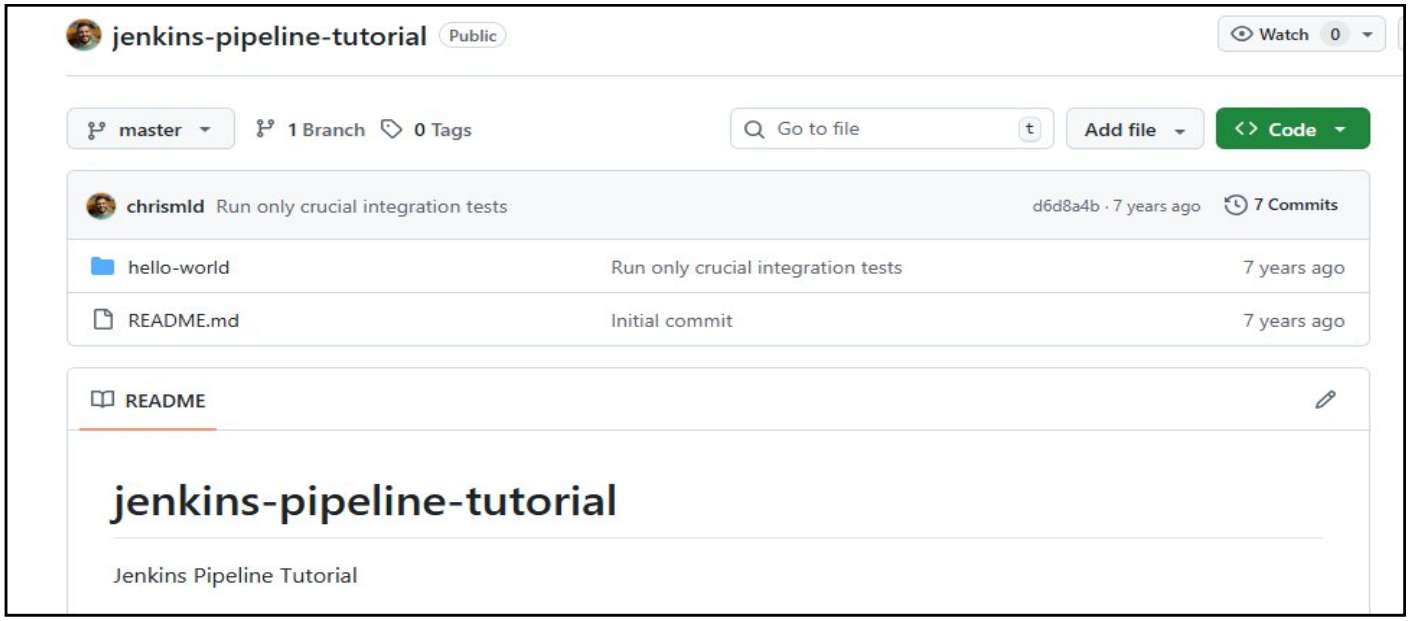


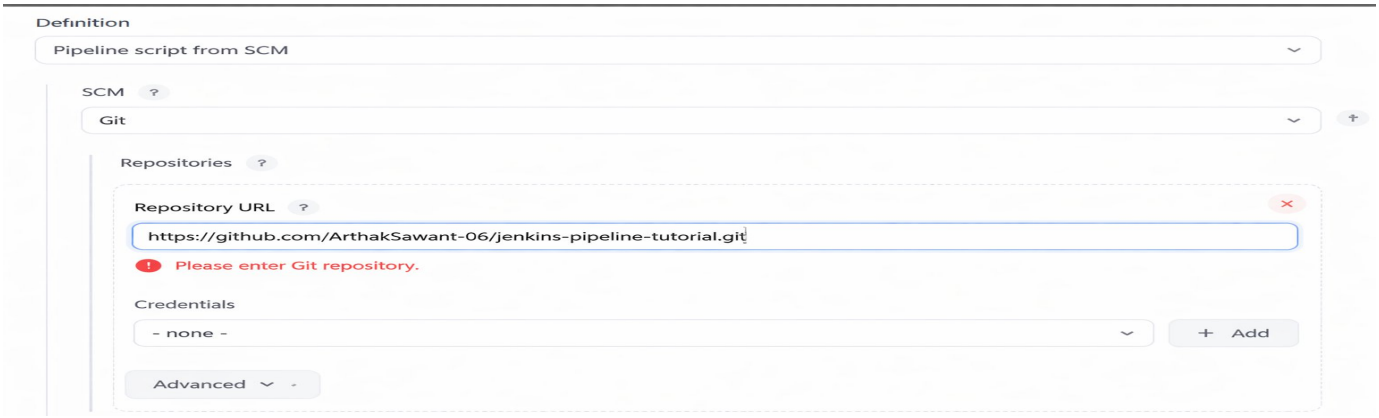

```
                    Integration test passed
                    [Pipeline] }
                    [Pipeline] // stage
                    [Pipeline] stage
                    [Pipeline] { (testing phase)
                    [Pipeline] parallel
                    [Pipeline] { (Branch: unit test)
                    [Pipeline] stage
                    [Pipeline] { (unit test)
                    [Pipeline] echo
[unit test]         running unit test
                    [Pipeline] }
                    [Pipeline] // stage
                    [Pipeline] }
                    [Pipeline] // parallel
                    [Pipeline] }
                    [Pipeline] // stage
                    [Pipeline] }
                    [Pipeline] // node
                    [Pipeline] End of Pipeline
                    Finished: SUCCESS
```

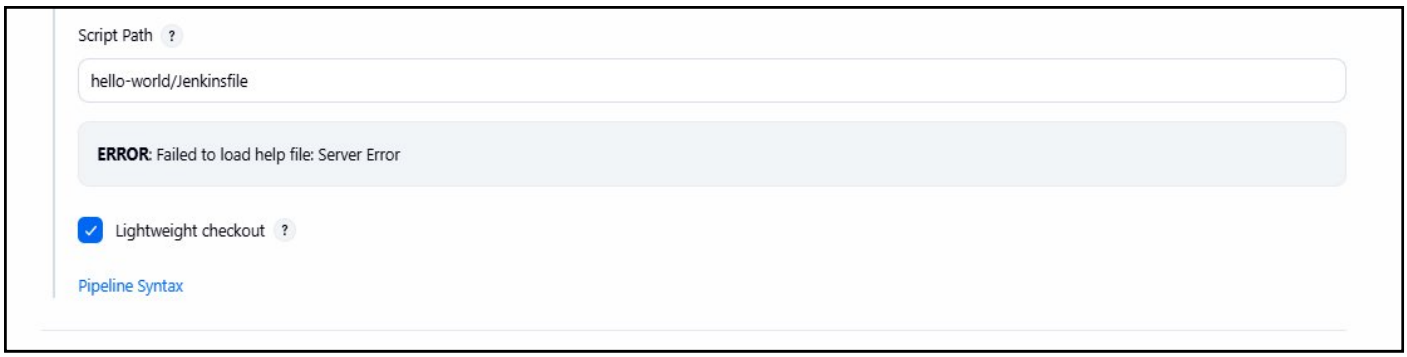


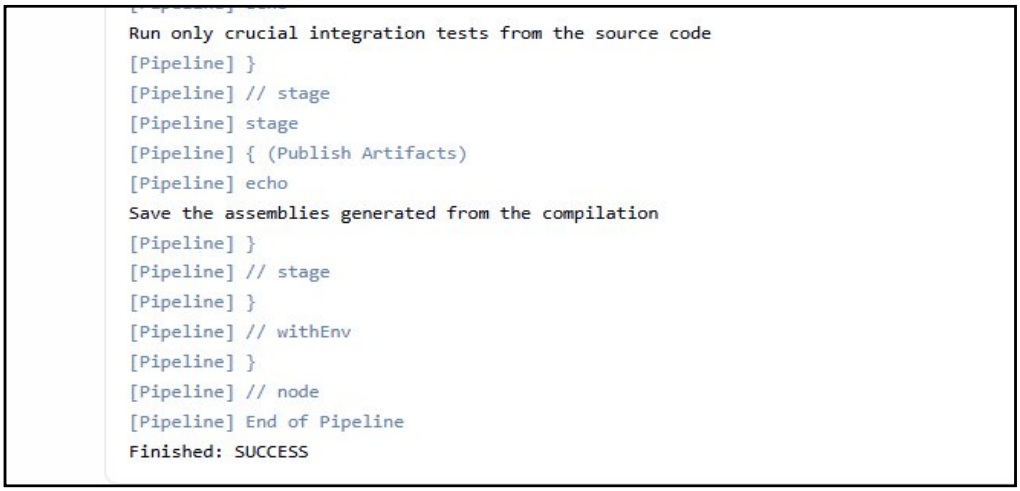


3. Pipeline Project with GitHub.

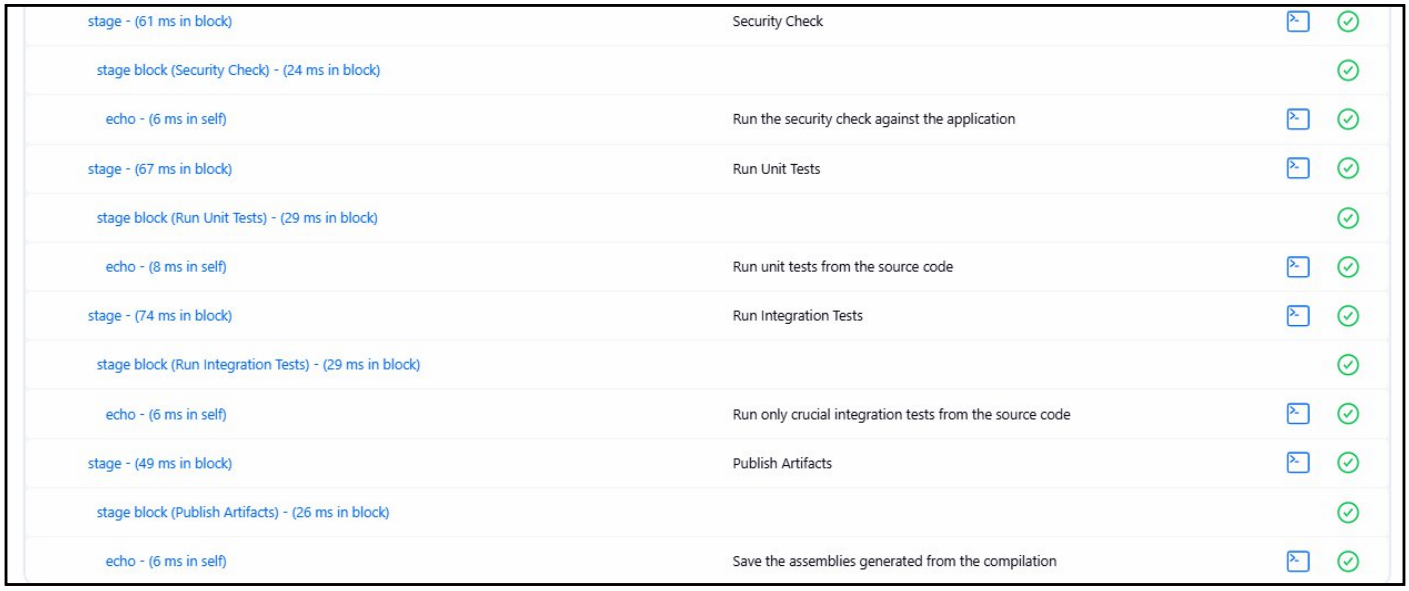



Definition
Pipeline script from SCM
SCM
Git
Repositories
Repository URL
https://github.com/ArthakSawant-06/jenkins-pipeline-tutorial.git
Please enter Git repository.
Credentials
- none -
Add
Advanced

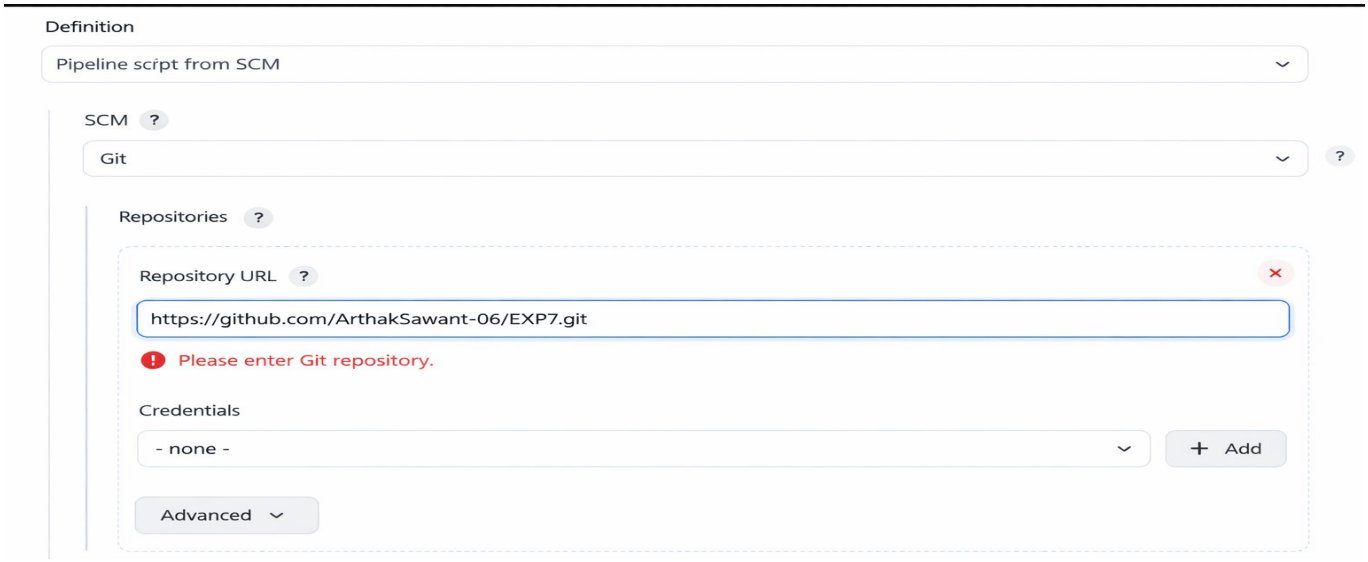

Script Path
hello-world/Jenkinsfile
ERROR: Failed to load help file: Server Error
Lightweight checkout
Pipeline Syntax


Run only crucial integration tests from the source code
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Publish Artifacts)
[Pipeline] echo
Save the assemblies generated from the compilation
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS

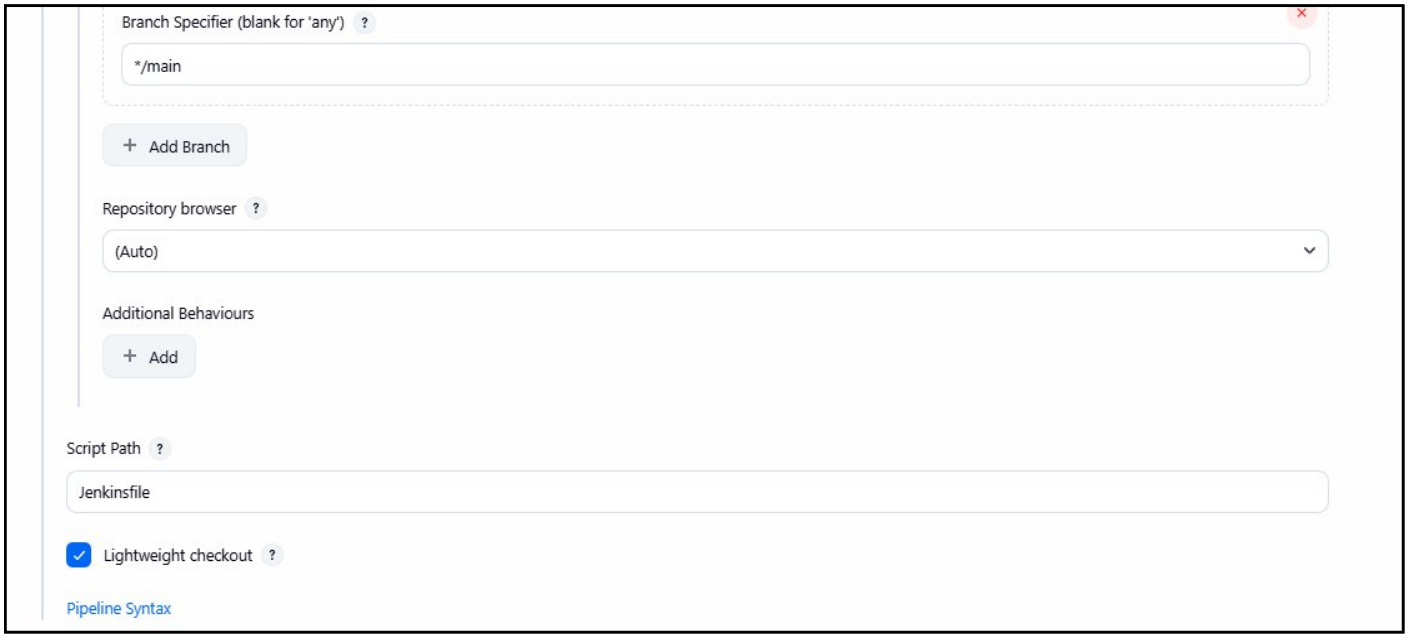

#1
Rerun
Started by admin
Started 42 sec ago
Queued 1 ms
Took 4.3 sec on built-in
Graph
Start
Checkout SCM
Static Analysis
Compile
Security Check
Run Unit Tests
Run Integration Tests
Publish Artifacts
End
Search
Checkout SCM 1.6s
Static Analysis 46ms
Compile 27ms
Security Check 24ms
Run Unit Tests 29ms
Run Integration Tests 29ms
Publish Artifacts 26ms
Publish Artifacts
26ms
Started 38s ago
Jenkins
Save the assemblies generated from the compilation
6ms
Save the assemblies generated from the compilation



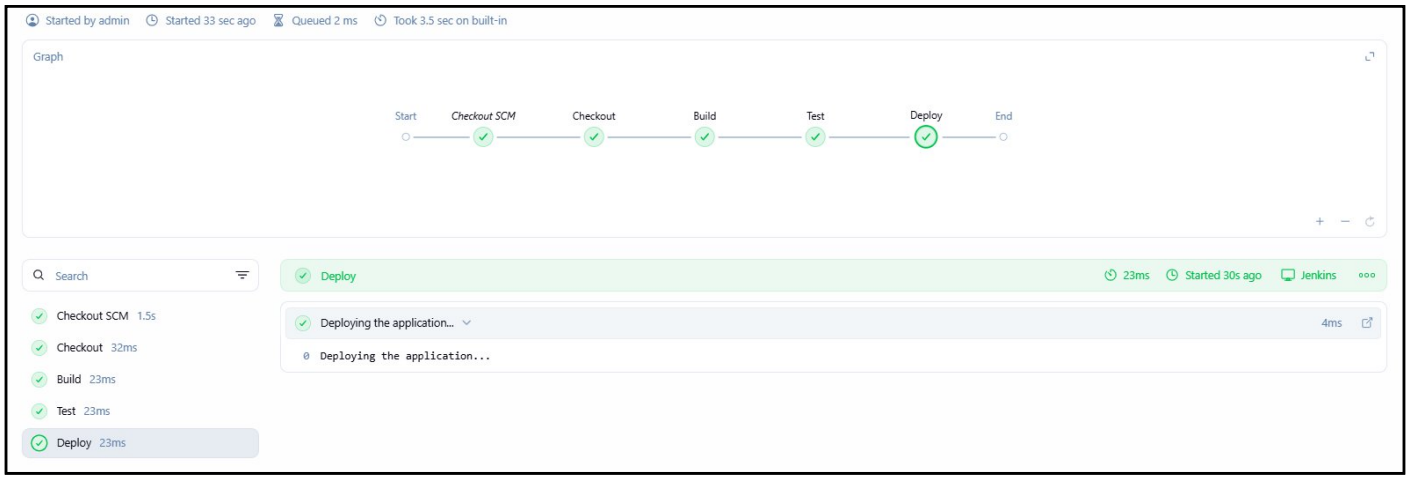
4. Create pipeline project with pipeline script from SCM

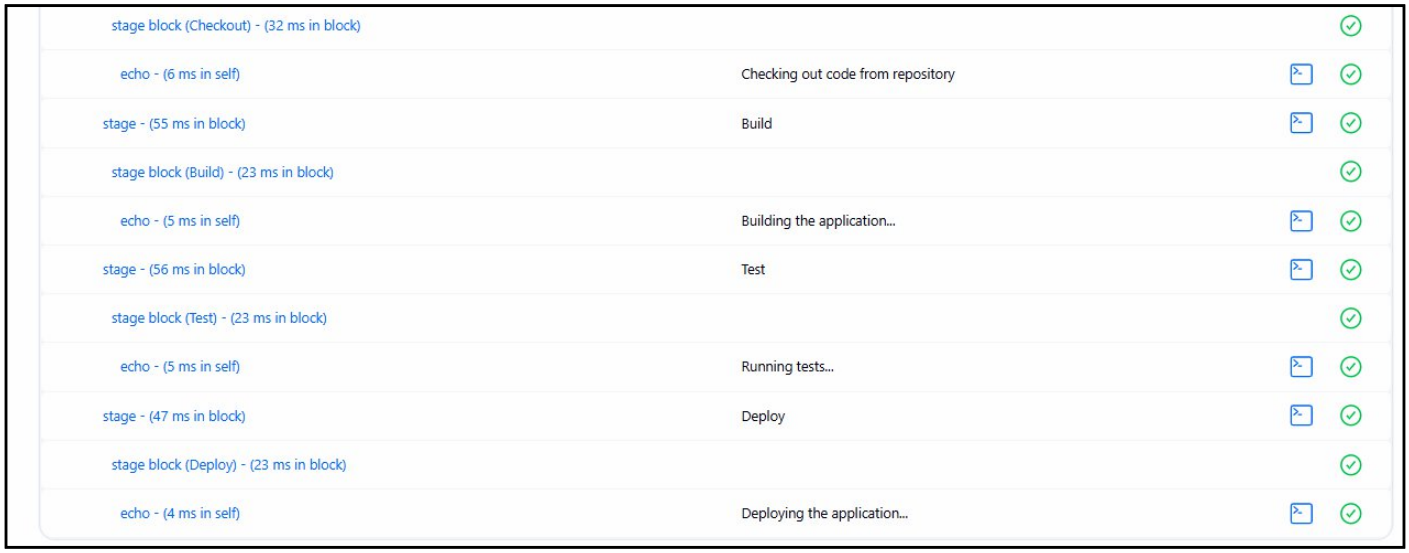


*/main is imp here



```
Building the application...
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Test)
[Pipeline] echo
Running tests...
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy)
[Pipeline] echo
Deploying the application...
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```





Pipeline Script

```
   pipeline {
agent any

stages {
   stage('Build') {
      steps {
         echo 'Building application...'
      }
}
```

```
      stage('Test') {
      steps {
         echo 'Running tests...'
      }
   }

   stage('Deploy') {
      steps {
         echo 'Deploying application...'
      }}}
```
